Information & examples

Lesson 1 · Introducing force and motion · Student notes

Why study motion?

Real car journeys can be complicated: vehicles accelerate, brake, corner, jump and collide. Scientists simplify real situations into models that help us describe, measure, communicate, predict, explain and ultimately control motion. In this first lesson, we focus on the language of motion, not calculations.

What is a physical quantity?

A physical quantity is something that can be measured or described quantitatively. A measurement normally includes a number and a unit, such as 12 metres. We use short symbols in science to represent quantities and units. These are not the same thing: v can represent velocity, while m/s is its SI unit.

Quantity	Typical symbol	SI unit (symbol)
Position	x	metre (m)
Distance travelled	d	metre (m)
Displacement	Δx	metre (m)
Speed	v or speed	metre per second (m/s)
Velocity	v (with direction)	metre per second (m/s)
Average speed	v_{avg} (speed)	metre per second (m/s)
Average velocity	$\bar{v}$ or $\Delta x/\Delta t$	metre per second (m/s)
Acceleration	a	metre per second squared (m/s^2)

Symbols vary between textbooks: the meaning must always be stated clearly. Δ means “change in”. Speed and velocity may share the letter v, but they are different ideas.

The eight quantities, in everyday language

1 Position Where the car is relative to a chosen reference point (origin). A car can be left or right of the origin.

2 Distance travelled How much road the car travels along its actual path, regardless of direction.

3 Displacement The overall change in the car’s position, from start to finish. It has a direction.

4 Speed How fast the car is moving at a particular moment, without specifying direction.

The eight quantities (continued)

5 Velocity How fast the car is moving AND in which direction. Turning changes velocity even if speed is constant.

6 Average speed How fast the car travels over the entire interval, based on its total distance travelled and elapsed time.

7 Average velocity The overall change in position, with direction, during the interval, relative to the elapsed time.

8 Acceleration How rapidly velocity changes. A car accelerates when it speeds up, slows down OR turns. Acceleration has a direction.

Scalars, vectors and direction

Scalar	Vector
Described by size (magnitude) only.	Described by magnitude AND direction.
Examples: distance, speed, average speed.	Examples: displacement, velocity, average velocity, acceleration.
“30 m/s” can describe speed.	“30 m/s east” can describe velocity.

An arrow can represent a vector: it points in the direction of the quantity, and its length can represent magnitude. In this unit we will mostly look at straight-line motion, so we can choose one direction as positive (+) and the opposite direction as negative (−). Negative velocity simply means motion in the negative direction, not “negative speed”.

Instantaneous vs average

At an instant: a snapshot	Over an interval: a whole journey
Position: where it is now.	Distance: total path covered.
Speed: what the speedometer reads now.	Displacement: change from start to finish.
Velocity: speed plus direction now.	Average speed: overall pace of travel.
Acceleration: how velocity is changing now.	Average velocity: overall change of position per elapsed time.

Car examples that matter

Out and back A car returns to where it started. Distance travelled is not zero, but displacement is zero. Its average speed is not zero, while its average velocity is zero.

At constant speed around a bend The car’s speed stays unchanged, but its velocity changes because direction changes; therefore it is accelerating.

Braking A car moving right while slowing has a leftward acceleration: acceleration does not always mean “going faster”.

Questions to keep asking

- What exactly am I describing: an instant or an interval?
- Is this quantity a scalar or a vector?
- What is the reference point and which direction is positive?